# Insurance Claims Analysis

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Project Link:

https://github.com/vanshag1208/Insurance-Portfolio-and-Claims-Analytics

Python Scripts:

https://github.com/vanshag1208/Insurance-Portfolio-and-Claims-Analytics/tree/main/python%20scripts

CSV Datasets:

https://github.com/vanshag1208/Insurance-Portfolio-and-Claims-Analytics/tree/main/data

SQL Queries:

https://github.com/vanshag1208/Insurance-Portfolio-and-Claims-Analytics/blob/main/sql/queries.sql

Power BI Dashboard:

https://github.com/vanshag1208/Insurance-Portfolio-and-Claims-Analytics/blob/main/power%20bi%20dashboard/analysis.pdf

Project Documentation:

https://github.com/vanshag1208/Insurance-Portfolio-and-Claims-Analytics/blob/main/reports%20and%20analysis/Insurance%20Claims%20Analysis%20Project%20Documentation.pdf

Analytical Questions:

https://github.com/vanshag1208/Insurance-Portfolio-and-Claims-Analytics/blob/main/reports%20and%20analysis/Analytical%20Questions.pdf

## Project Documentation

### 1. Project Objective

The objective of this project is to analyze an insurance portfolio using simulated policy and claims data. The analysis focuses on understanding premium collection, claim patterns, policy tenure distribution, and overall portfolio profitability using business metrics such as loss ratio and claim rate.

The goal is to derive insights that can help an insurance company evaluate the performance of its portfolio and identify risk patterns across different policy tenures.

### 2. Dataset Overview

The dataset consists of two main tables.

**Policy Sales Table**

This table contains policy purchase information and policy details.

Columns include:

- Customer_ID
- Vehicle_ID
- Vehicle_Value
- Premium
- Policy_Purchase_Date
- Policy_Start_Date
- Policy_End_Date
- Policy_Tenure

Each row represents a policy purchased by a customer.

**Claims Table**

This table records insurance claims made against policies.

Columns include:

- Claim_ID
- Customer_ID
- Vehicle_ID
- Claim_Amount
- Claim_Date
- Claim_Type

Each row represents a claim made on a specific policy.

**3. Project Workflow**

The project followed an end-to-end data analytics workflow.

**Step 1 – Data Generation**

Python was used to simulate policy and claims data based on the rules defined in the assignment.

The following business rules were implemented:

- Vehicle value fixed at ₹100,000
- Claim amount set to 10% of vehicle value (₹10,000)
- Premium calculated as ₹100 per year of policy tenure
- Policy tenures distributed across 1–4 years
- Claims generated for both 2025 and 2026

The generated datasets were exported as CSV files.

**Step 2 – Data Storage**

The generated datasets were imported into **SQL Server Management Studio (SSMS)** for structured storage and validation.

Tables created:

- policy_sales
- claims

SQL was used to verify:

- missing values
- date formats
- row counts
- basic data integrity

**Step 3 – Data Cleaning**

During the data loading process, a few issues were identified:

**Issue 1 – Date Format Problem**

Some Claim_Date values were not correctly interpreted during the SQL import due to datetime formatting differences.

Resolution:

- The dataset was corrected and re-imported to ensure proper date parsing.

**Issue 2 – Metric Interpretation**

Initially, the very high loss ratio appeared incorrect. However, after reviewing the dataset assumptions, it was determined that the result was expected because the claim amount is significantly larger than the premium collected.

**Step 4 – Data Modeling**

In Power BI, a relationship was created between the two tables:

policy_sales.Vehicle_ID → claims.Vehicle_ID

Relationship type:

One-to-Many

This enabled policy attributes such as tenure and premium to be analyzed alongside claims data.

**4. Dashboard Development**

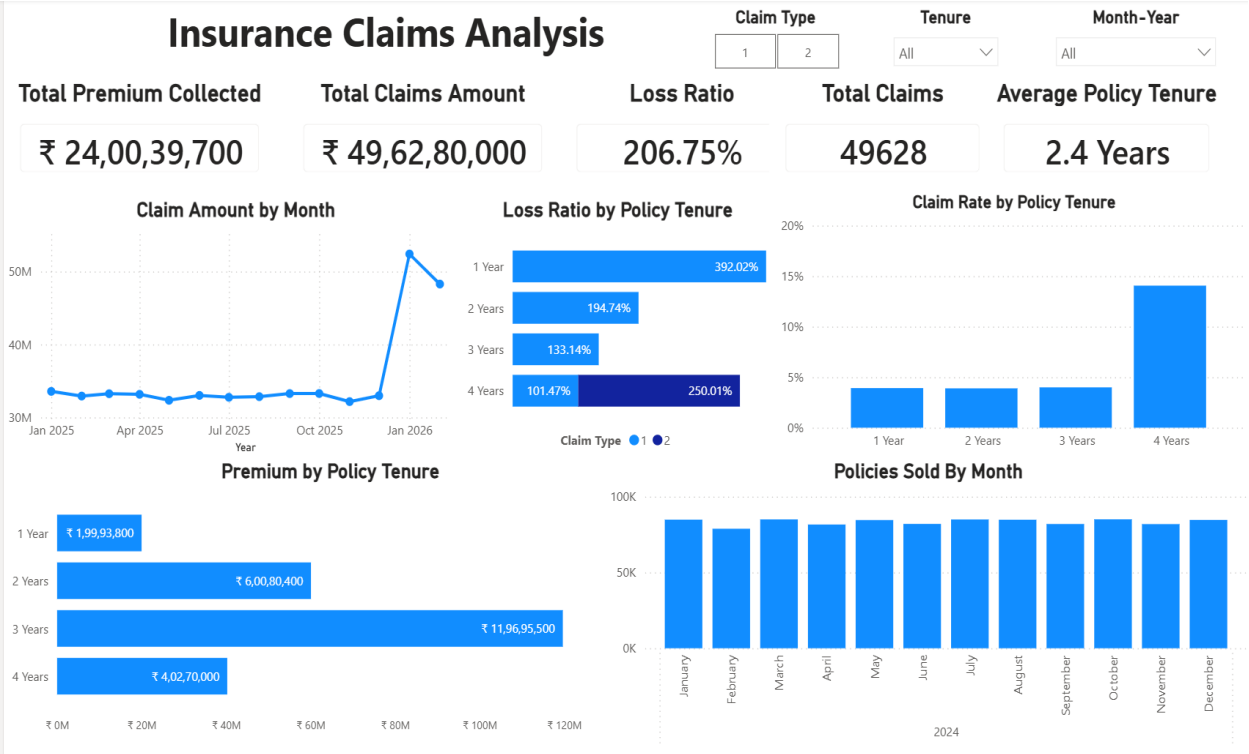
A Power BI dashboard was built to visualize key insurance metrics and trends.

The dashboard includes the following KPIs:

- Total Premium Collected
- Total Claims Amount
- Loss Ratio
- Total Claims Count
- Average Policy Tenure

Additional visuals were included to analyze trends and portfolio performance.

**5. Dashboard Visuals**




**Claim Amount by Month**

This visual shows how claim amounts vary across months.

It helps identify periods with unusually high claim activity.

**Premium by Policy Tenure**

This chart displays the distribution of premium revenue across policy tenures.

It shows which tenure contributes the most revenue to the portfolio.

**Loss Ratio by Policy Tenure**

Loss ratio measures the profitability of the insurance portfolio.

Formula:

Loss Ratio = Total Claims / Total Premium

These visual highlights which policy tenures generate higher financial risk.

**Claim Rate by Policy Tenure**

Claim rate measures how frequently customers make claims.

Formula:

Claim Rate = Number of Claims / Total Policies

This helps analyse customer behaviour across different policy durations.

**Policies Sold by Month**

These visual tracks policy sales trends across different months.

It helps understand portfolio growth patterns.

**6. Key Insights**

Several insights were derived from the dashboard.

**Insight 1 – Portfolio Loss**

The portfolio shows a **loss ratio of approximately 206.75%**, meaning total claims exceed total premiums collected.

This occurs because the simulated claim amount (₹10,000) is significantly larger than the premium charged.

**Insight 2 – Short Tenure Policies Are Riskier**

1-year policies have the highest loss ratio.

This is expected because the premium collected for short tenures is relatively small compared to the claim amount.

**Insight 3 – Three-Year Policies Perform Best**

Policies with a tenure of three years show the lowest loss ratio, indicating relatively better portfolio performance.

**Insight 4 – Claim Frequency in Long Tenures**

Policies with longer durations tend to have a higher claim rate, indicating that customers holding policies for longer periods are more likely to make claims.

**Insight 5 – Stable Policy Sales**

Policy sales appear relatively consistent across months, suggesting steady portfolio growth.

## 7. Use of AI Assistance

Artificial intelligence tools were used during the project primarily to:

- speed up development
- validate analytical logic
- refine documentation

AI assistance was used to help with:

- structuring SQL queries
- improving Power BI measure definitions
- organizing project documentation

However, the dataset generation, dashboard design decisions, analysis logic, and interpretation of results were performed independently.

AI was used as a productivity tool rather than as a replacement for analytical reasoning.

## 8. Conclusion

This project demonstrates an end-to-end data analytics workflow involving:

- synthetic data generation
- SQL data validation
- Power BI dashboard development
- insurance portfolio analysis

The analysis highlights how policy tenure impacts portfolio risk and claim behavior.

Such insights can help insurance companies refine pricing strategies and better understand the risk profile of their policyholders.